

***B. Tech. Degree VI Semester Special Supplementary
Examination November 2013***

**ME 602 DYNAMICS OF MACHINERY
(2006 Scheme)**

Time : 3 Hours

Maximum Marks : 100

PART A
(Answer *ALL* questions)

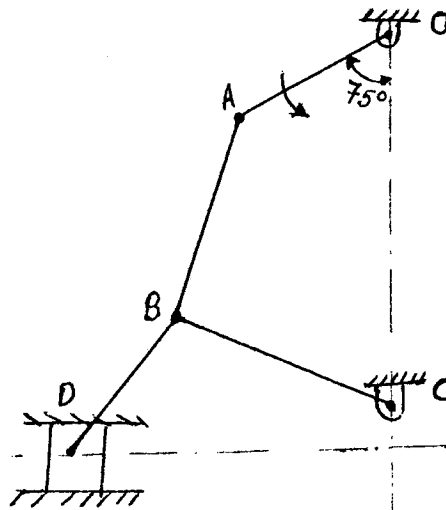
(8 x 5 = 40)

- I. (a) State and explain D' Alembert's principle.
(b) Sketch and describe the four bar chain mechanism.
(c) Explain the terms 'fluctuation of energy' and 'fluctuation of speed' as applied to flywheel.
(d) Define and explain the following terms relating to governors.
(i) Stability (ii) Isochronism (iii) Hunting
(e) Explain the terms static balancing and dynamic balancing. State the necessary conditions to achieve them.
(f) Explain the terms swaying couple and hammer blow with suitable figures.
(g) Discuss relative merits and demerits of belt and rope drive for the power transmission.
(h) Distinguish between brakes and dynamometer.

PART B

(4 x 15 = 60)

- II. In the figure, the angular velocity of the crank OA is 600 r.p.m. Determine the linear velocity of the slider D and the angular velocity of the link BD, when the crank is inclined at an angle of 75° to the vertical. The dimensions of the various links are: OA=28mm; AB=44mm; BC=49mm; and BD=46mm. The center distance between the centers of rotation O and C is 65mm. The path of travel of the slider is 11mm below the fixed point C. The slider moves along a horizontal path and OC is vertical.



OR

(P.T.O.).

- III. During a trial on steam engine, it is found that the acceleration of the piston is 36m/s^2 when the crank has moved 30° from the inner dead center position. The net effective steam pressure on the piston is 0.5 N/mm^2 and the frictional resistance is equivalent to a force of 600N . The diameter of the piston is 300mm and the mass of the reciprocating parts is 180kg . If the length of the crank is 300mm and the ratio of the connecting rod length to the crank length is 4.5 , find 1. Reaction on the guide bars, 2. Thrust on the crank shaft bearings, and 3. Turning moment on the crank shaft.

- IV. The turning moment diagram for a multicylinder engine has been drawn to a scale $1\text{mm} = 600\text{N-m}$ vertically and $1\text{mm} = 3^\circ$ horizontally. The intercepted areas between the output torque curve and the mean resistance line, taken in order from one end are as follows: $+52, -124, +92, -140, +85, -72$ and $+107\text{mm}^2$, when the engine is running at a speed of 600 r.p.m. If the total fluctuation of speed is not to exceed $\pm 1.5\%$ of the mean, find the necessary mass of the flywheel of radius 0.5m .

OR

- V. The mass of the turbine rotor of the ship is 20tonnes and has a radius of gyration of 0.6m . Its speed is 2000 r.p.m. the ship pitches 6° above and 6° below the horizontal position. A complete oscillation takes 30 seconds and the motion is simple harmonic. Determine 1. Maximum gyroscopic couple, 2. Maximum angular acceleration of the ship during pitching and 3. The direction in which the bow will tend to turn when rising, if the rotation of the rotor is clockwise when looking from the left.

- VI. Four masses m_1, m_2, m_3 and m_4 are $200\text{kg}, 300\text{kg}, 240\text{kg}$ and 260kg respectively. The corresponding radii of rotation are $0.2\text{m}, 0.15\text{m}, 0.25\text{m}$ and 0.3m respectively and the angles between successive masses are $45^\circ, 75^\circ$ and 135° . Find the position and magnitude of the balance mass required, if the radius of rotation is 0.2m .

OR

- VII. An inside cylinder locomotive has its cylinder centre lines 0.7m apart and has a stroke of 0.6m . The rotating masses per cylinder are equivalent to 150kg at the crank pin, and the reciprocating masses per cylinder to 180kg . The wheel centre lines are 1.5m apart. The crank is at right angles. The whole of the rotating and $2/3$ of the reciprocating masses are to be balanced by masses placed at a radius of 0.6m . Find the magnitude and direction of the balancing masses. Also find the fluctuation in rail pressure under one wheel, variation of tractive effort and the magnitude of swaying couple at a crank speed of 300 r.p.m.

- VIII. Two parallel shafts whose centre lines are 4.8m apart, are connected by open belt drive. The diameter of the larger pulley is 1.5m and that of the smaller pulley 1m . The initial tension in the belt when stationary is 3kN . The mass of the belt is 1.5kg/m length. The coefficient of friction between the belt and the pulley is 0.3 . Taking centrifugal tension into account, calculate the power transmitted, when the smaller pulley rotates at 400 r.p.m.

OR

- IX. Describe with a neat sketch the working of a single plate friction clutch and deduce an equation for the total frictional torque transmitted when there is a uniform pressure.
